NEXVIA INTERNSHIP 2026 • FINAL CAPSTONE PRESENTATION

Large-Scale Personalized Recommendation Engine

A Two-Stage Retrieval & Deep Ranking Architecture Modeled After Netflix & Amazon

Presenter: Bisma Mustafa **Dataset:** MovieLens 25M (11,641 Indexed Movies) **Tech Stack:** TensorFlow, FAISS, PySpark, FastAPI, Redis, Docker, Airflow

Slide 1: Problem Statement & Motivation

- **The Scale vs. Latency Bottleneck:** Modern streaming and e-commerce platforms must score millions of candidate items per user in under 100 milliseconds.
- **Why Single-Stage Models Fail:** Evaluating every item with a deep neural network causes prohibitive inference latency and compute bottlenecks.
- **The Two-Stage Solution:** Decouple fast candidate retrieval (Stage 1) from high-precision deep ranking (Stage 2) to balance recall and accuracy.



Two-Stage Neural Retrieval & Ranking Concept

Slide 2: End-to-End Two-Stage System Architecture

SYSTEM DESIGN & DATA FLOW

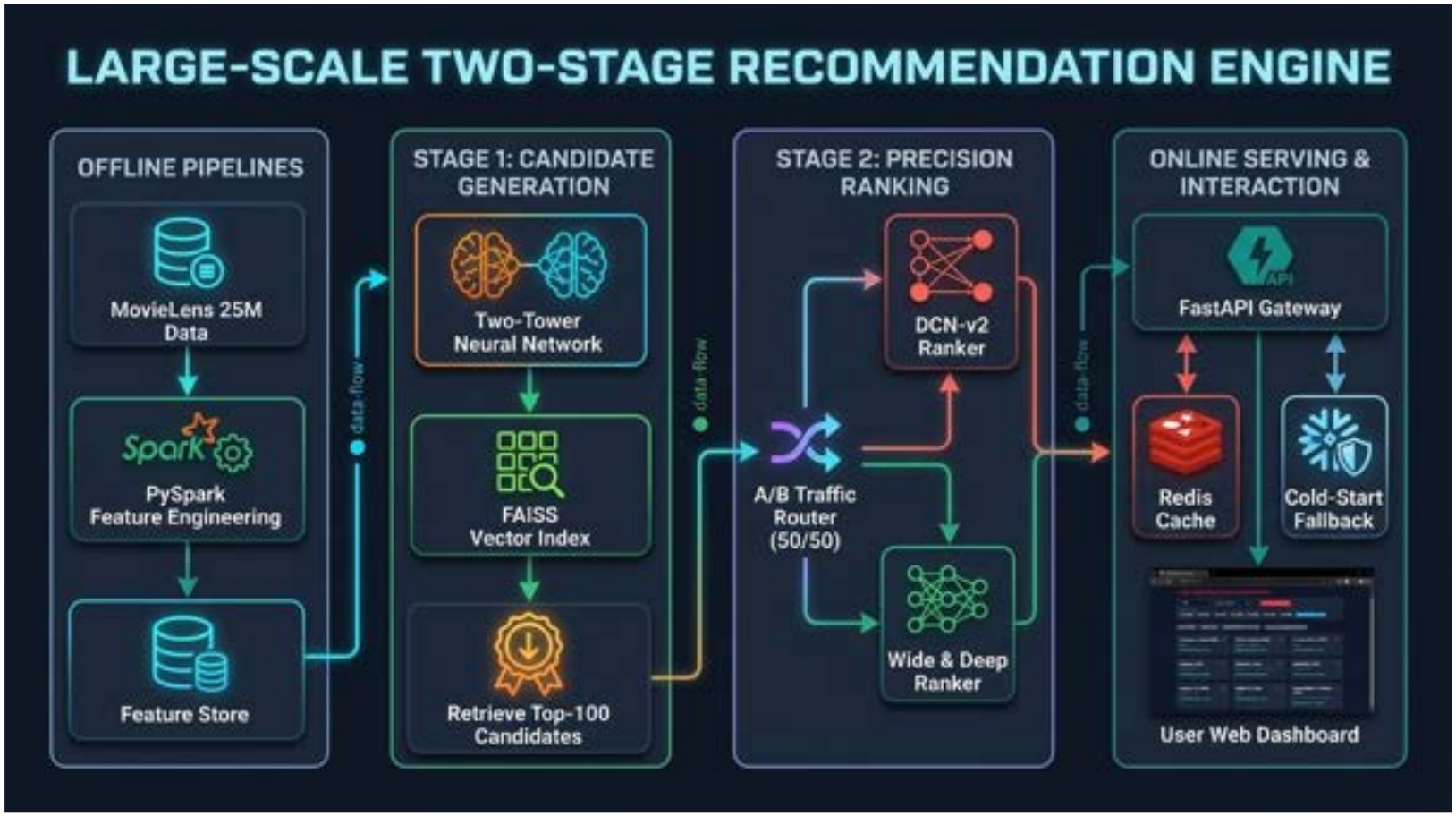


Figure 1: Offline PySpark Feature Pipelines, Stage-1 Two-Tower + FAISS Retrieval, Stage-2 DCN/Wide & Deep A/B Ranking, and FastAPI Serving.

Slide 3: Engineering Challenges & Architectural Solutions

DATA INTEGRITY & PRODUCTION RESILIENCE

Engineering Challenge	Impact if Unaddressed	Implemented Solution
Future Data Leakage	Random train/test splits allow models to peek at future ratings, inflating offline metrics artificially.	Strict global chronological temporal split (80% Train / 10% Val / 10% Test) with shifted rolling user windows.
Cold-Start Users & Items	New users lack interaction embeddings, causing collaborative filtering models to fail or output noise.	Bayesian-damped weighted rating fallbacks and genre-specific top-K lists cached in JSON/Redis.
Sub-Millisecond Retrieval	Brute-force dot-product comparisons across tens of thousands of items degrade API response times.	64-dimensional L2-normalized Two-Tower embeddings indexed in FAISS (`IndexIDMap`) for instant top-100 recall.
Continuous Model Drift	User preferences evolve over time, making static recommendation models stale.	Automated weekly Apache Airflow DAG orchestrating feature extraction, retraining, FAISS rebuilds, and Triton updates.

Slide 4: Offline Evaluation & Online A/B Test Results

PERFORMANCE BENCHMARKS

We evaluated Stage-1 retrieval quality and Stage-2 ranking precision across the chronological validation split, followed by a 50/50 deterministic MD5 traffic split comparing **DCN-v2 (Treatment)** against **Wide & Deep (Control)**.

Model Variant	AUC	Recall@20	NDCG@20	Online CTR
DCN-v2 Ranker	0.8177	0.0461	**0.0584**	**14.10% (+13.71%)**
Wide & Deep	0.8178	0.0461	0.0521	12.40% (Control)

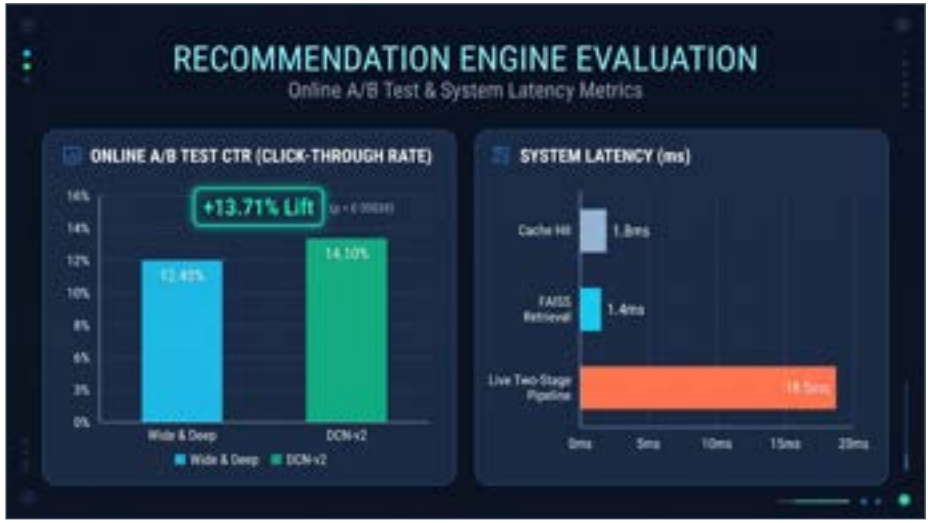


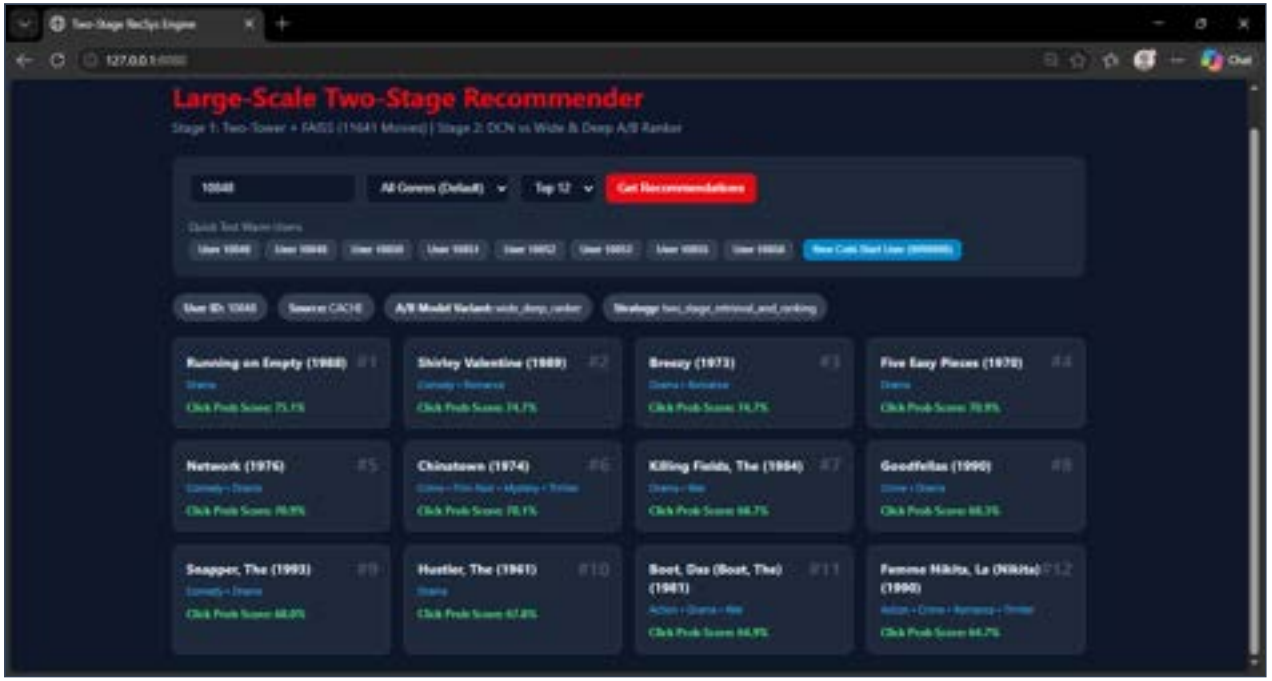
Figure 2: Online A/B Test CTR Lift (+13.71%) and Sub-20ms End-to-End Serving Latency Benchmarks.

Statistical Significance & Key Takeaways

- **Two-Proportion Z-Test:** Across 20,000 simulated impressions, DCN-v2 achieved **Z = 3.55** (p = 0.00039), confirming statistically significant CTR lift.
- **Feature Crosses:** Explicit cross layers captured interactions between rolling user ratings and item popularity more effectively than linear wide layers.

Slide 5: Live FastAPI Dashboard & Inference Walkthrough

PRODUCTION SERVING LAYER



Live FastAPI Dashboard serving ranked top-12 recommendations for User 10848 via Wide & Deep Ranker with active caching.

Live System Capabilities

- **Cache-First Latency:** Subsequent requests for User 10848 serve directly from CACHE without re-invoking neural models.
- **A/B Variant Routing:** Deterministic hashing assigns users consistently to either `dcn_ranker` or `wide_deep_ranker`.
- **Side-by-Side Comparison:** Dedicated `/compare/{user_id}` endpoint scores the same 100 FAISS candidates across both rankers simultaneously.
- **Cold-Start Safety:** Unknown users (e.g., ID 9999999) seamlessly receive Bayesian popular or genre-filtered recommendations.

Slide 6: Next Steps & Individual Intern Reflection

ROADMAP & OWNERSHIP

Future Roadmap & Production Next Steps

- **Real-Time Event Streaming:** Integrate Apache Kafka to capture live user clicks and update user embeddings in near real-time rather than weekly batch windows.
- **Multi-Objective Optimization:** Extend Stage-2 ranking heads to jointly optimize click probability, watch-time/rating value, and diversity penalties.

Individual Intern Reflection — Bisma Mustafa

What I Learned: Building this two-stage recommender gave me practical mastery over how large-scale platforms balance retrieval speed with ranking precision. I learned how to prevent temporal data leakage, train contrastive Two-Tower embeddings in TensorFlow, index vectors in FAISS, and validate model improvements rigorously using NDCG@K, Recall@K, and statistical Z-tests.

What I Owned: I took full end-to-end ownership of the pipeline—from chronological dataset preprocessing and rolling feature engineering to training both DCN-v2 and Wide & Deep rankers, writing Triton/Airflow/Docker deployment configurations, and building the interactive FastAPI dashboard for live A/B testing.